TP MODUL 5

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Kelas : IT-08-02

DIGIT 9 NIM = 0 maka Sisa 0 = data yang disimpan adalah integer

1. main.cpp




```cpp
#include "list.h"

int main() {
    List L;
    createList_108092500012(L);
    int menu;
    char kembali;

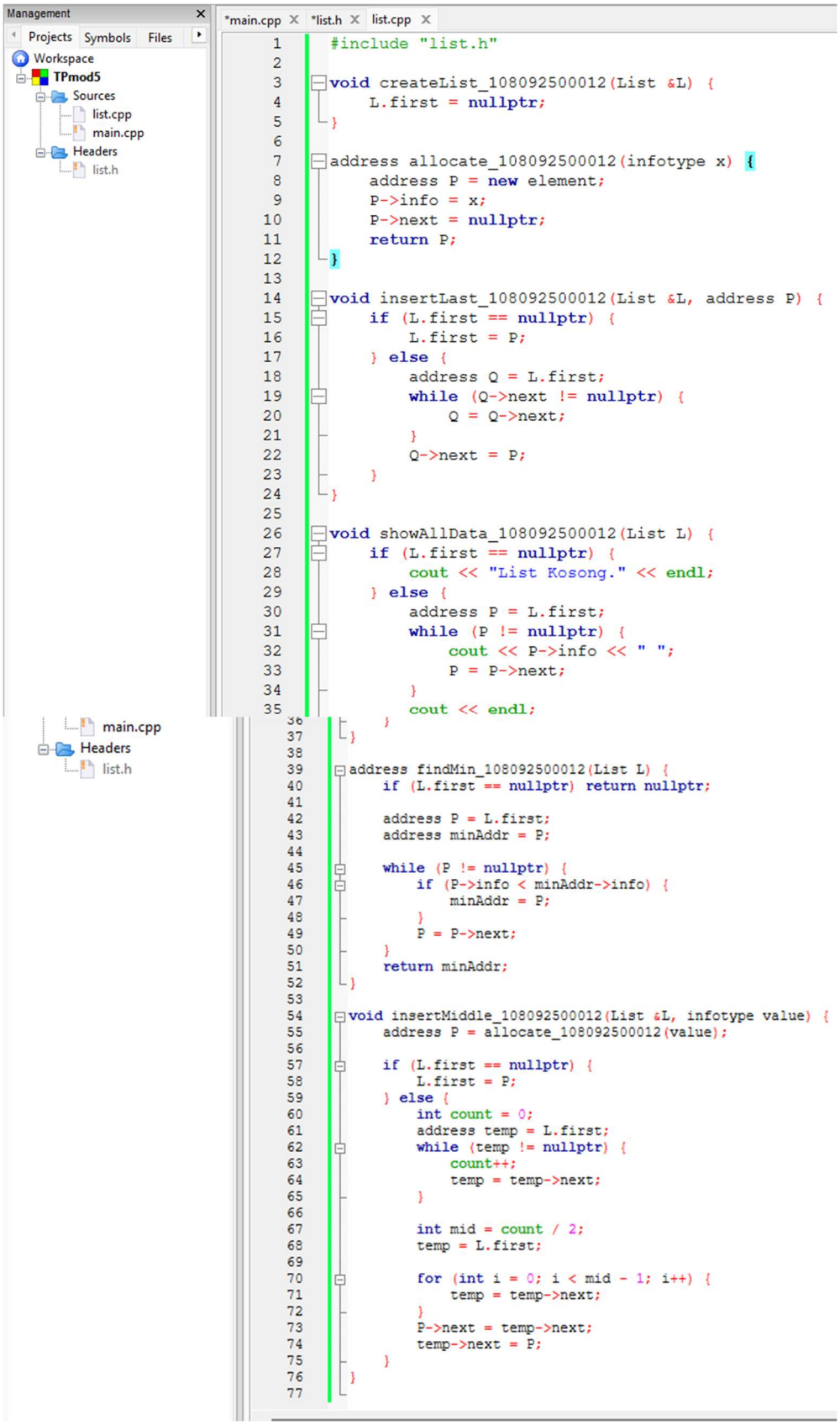
    do {
        cout << "======MENU=======\n";
        cout << "1. Menambah N data baru\n";
        cout << "2. Menampilkan semua data\n";
        cout << "3. Cari alamat nilai terkecil (findMin)\n";
        cout << "4. Sisip data di tengah (insertMiddle)\n";
        cout << "0. Exit\n";
        cout << "Masukkan menu: ";
        cin >> menu;

        if (menu == 1) {
            int n, data;
            cout << "Jumlah data yang akan ditambahkan: ";
            cin >> n;
            for (int i = 0; i < n; i++) {
                cout << "Masukkan data baru: ";
                cin >> data;
                address P = allocate_108092500012(data);
                insertLast_108092500012(L, P);
            }
        } else if (menu == 2) {
            cout << "Data list saat ini: ";
            showAllData_108092500012(L);
        } else if (menu == 3) {
            address minData = findMin_108092500012(L);
            if (minData != nullptr) {
                cout << "Alamat nilai terkecil: " << minData << "\n";
                cout << "Nilai terkecilnya adalah: " << minData->info << "\n";
            } else {
                cout << "List kosong.\n";
            }
        } else if (menu == 4) {
            int data;
            cout << "Masukkan data bilangan ke tengah: ";
            cin >> data;
            insertMiddle_108092500012(L, data);
            cout << "Data berhasil disisipkan.\n";
        } else if (menu == 0) {
            cout << "ANDA TELAH KELUAR DARI PROGRAM\n";
            break;
        } else {
            cout << "Pilihan tidak valid.\n";
        }

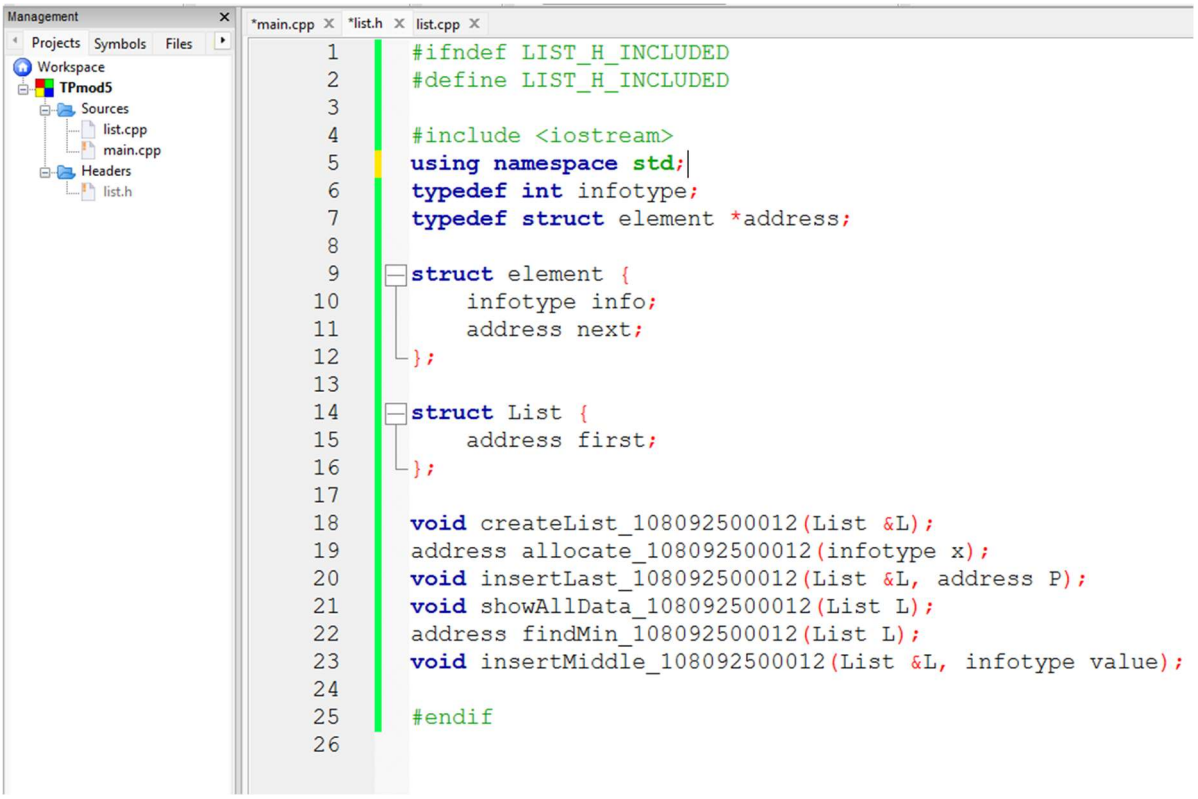
        if (menu != 0) {
            cout << "Kembali ke menu utama? (Y/N): ";
            cin >> kembali;
            if (kembali == 'N' || kembali == 'n') {
                cout << "ANDA TELAH KELUAR DARI PROGRAM\n";
                break;
            }
            cout << "\n";
        }
    } while (menu != 0);

    return 0;
}S
```

2. list.cpp


Management
Projects
Symbols
Files
Workspace
TPmod5
Sources
list.cpp
main.cpp
Headers
list.h
*main.cpp
*list.h
list.cpp
#include "list.h"
void createList_108092500012(List &L) {
L.first = nullptr;
}
address allocate_108092500012(infotype x) {
address P = new element;
P->info = x;
P->next = nullptr;
return P;
}
void insertLast_108092500012(List &L, address P) {
if (L.first == nullptr) {
L.first = P;
} else {
address Q = L.first;
while (Q->next != nullptr) {
Q = Q->next;
}
Q->next = P;
}
}
void showAllData_108092500012(List L) {
if (L.first == nullptr) {
cout << "List Kosong." << endl;
} else {
address P = L.first;
while (P != nullptr) {
cout << P->info << " ";
P = P->next;
}
cout << endl;
}
}
address findMin_108092500012(List L) {
if (L.first == nullptr) return nullptr;
address P = L.first;
address minAddr = P;
while (P != nullptr) {
if (P->info < minAddr->info) {
minAddr = P;
}
P = P->next;
}
return minAddr;
}
void insertMiddle_108092500012(List &L, infotype value) {
address P = allocate_108092500012(value);
if (L.first == nullptr) {
L.first = P;
} else {
int count = 0;
address temp = L.first;
while (temp != nullptr) {
count++;
temp = temp->next;
}
int mid = count / 2;
temp = L.first;
for (int i = 0; i < mid - 1; i++) {
temp = temp->next;
}
P->next = temp->next;
temp->next = P;
}
}

3. list.h


Management
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TPmod5
Sources
list.cpp
main.cpp
Headers
list.h
*main.cpp
*list.h
list.cpp
#ifndef LIST_H_INCLUDED
#define LIST_H_INCLUDED
#include <iostream>
using namespace std;
typedef int infotype;
typedef struct element *address;
struct element {
infotype info;
address next;
};
struct List {
address first;
};
void createList_108092500012(List &L);
address allocate_108092500012(infotype x);
void insertLast_108092500012(List &L, address P);
void showAllData_108092500012(List L);
address findMin_108092500012(List L);
void insertMiddle_108092500012(List &L, infotype value);
#endif

RUN & GO

```
Masukkan menu: 1
Jumlah data yang akan ditambahkan: 5
Masukkan data baru: 1
Masukkan data baru: 3
Masukkan data baru: 5
Masukkan data baru: 7
Masukkan data baru: 9
Kembali ke menu utama? (Y/N): y

======MENU=======
1. Menambah N data baru
2. Menampilkan semua data
3. Cari alamat nilai terkecil (findMin)
4. Sisip data di tengah (insertMiddle)
0. Exit
Masukkan menu: 2
Data list saat ini: 1 3 5 7 9
Kembali ke menu utama? (Y/N): y

======MENU=======
1. Menambah N data baru
2. Menampilkan semua data
3. Cari alamat nilai terkecil (findMin)
4. Sisip data di tengah (insertMiddle)
0. Exit
Masukkan menu: 3
Alamat nilai terkecil: 0x1c7af9e3380
Nilai terkecilnya adalah: 1
Kembali ke menu utama? (Y/N): y

======MENU=======
1. Menambah N data baru
2. Menampilkan semua data
3. Cari alamat nilai terkecil (findMin)
4. Sisip data di tengah (insertMiddle)
0. Exit
Masukkan menu: 4
Masukkan data bilangan ke tengah: 67
Data berhasil disisipkan.
Kembali ke menu utama? (Y/N): y

======MENU=======
1. Menambah N data baru
2. Menampilkan semua data
3. Cari alamat nilai terkecil (findMin)
4. Sisip data di tengah (insertMiddle)
0. Exit
Masukkan menu: 2
Data list saat ini: 1 3 67 5 7 9
Kembali ke menu utama? (Y/N): y

======MENU=======
1. Menambah N data baru
2. Menampilkan semua data
3. Cari alamat nilai terkecil (findMin)
4. Sisip data di tengah (insertMiddle)
0. Exit
Masukkan menu: 0
ANDA TELAH KELUAR DARI PROGRAM

Process returned 0 (0x0)   execution time : 24.635 s
Press any key to continue.
```